SUPPLY VS DEMAND - COPPER

Supply

- **Global Copper Production Trends (2003–2023): (1)**
  - **Production Growth (2003–2023)**: Global mine output up 65% (15.1M → 24.8M tons).
    - DR Congo: 70K → 3M tons (+4257%).
    - China: 800K → 1.9M tons (+138%).
    - Peru: 929K → 3,037K (+227%), overtaking the U.S. in production.
    - U.S. output remained flat (~1.2–1.5 million tons), losing global market share.
    - India - Copper ore production declined by approximately **19.55%** from 2018–19 to 2022–23. (2)
- **China's Made in China 2025 Initiative**: (3)
  - Expected to add 232,000 tons per year of copper demand by 2025 due to higher efficiency motor standards, increasing copper intensity in motors from 0.87 kg/kW in 2015 to 1.56 kg/kW in 2025.
- **U.S. Copper Supply Chain (2003–2023):**(1)
  - **Flat Mining**: 1.2–1.5M tons since 2000; Arizona ~⅔ output.
  - **Declining Processing**: Smelting 680K → 417K tons (2007–2023) = Reduced Domestic processing capacity; refined output 1,440K → 971K(2003-2023) short tons = Shrinking Refining capacity.
  - **Low Inventories**: Stocks down 723K → 140K short tons (80% drawdown).
  - **High Imports**: 600–1,000K short tons/year.
  - **Declining Electrowon Production:** 652K → 568K short tons (2003–2023) = 12.88% decline
  - **Low scrap contribution:** Recycling adds only 40–60 thousand short tons annually, insufficient to offset primary demand.
- **Chilean Copper Production (2004–2023)**:(4)
  - Between 2004–2013, Chile's copper production averaged 5,386.6 thousand metric tons (kMT) annually, with modest growth of 6.7% (5,412.5 kMT to 5,776.0 kMT). Codelco's output declined by 2.6%, while private producers like Anglo American Sur grew by 55.5%.
  - Refined copper production in Chile fell by 15.9% after peaking in 2009, indicating declining refining capacity.
  - From 2014–2023, average copper ore grades in Chile declined: concentrator plants from 0.90% to 0.69%, heap leaching from 0.67% to

0.44%, and national weighted average from 0.72% to 0.58%, signaling ore quality depletion and higher extraction costs.

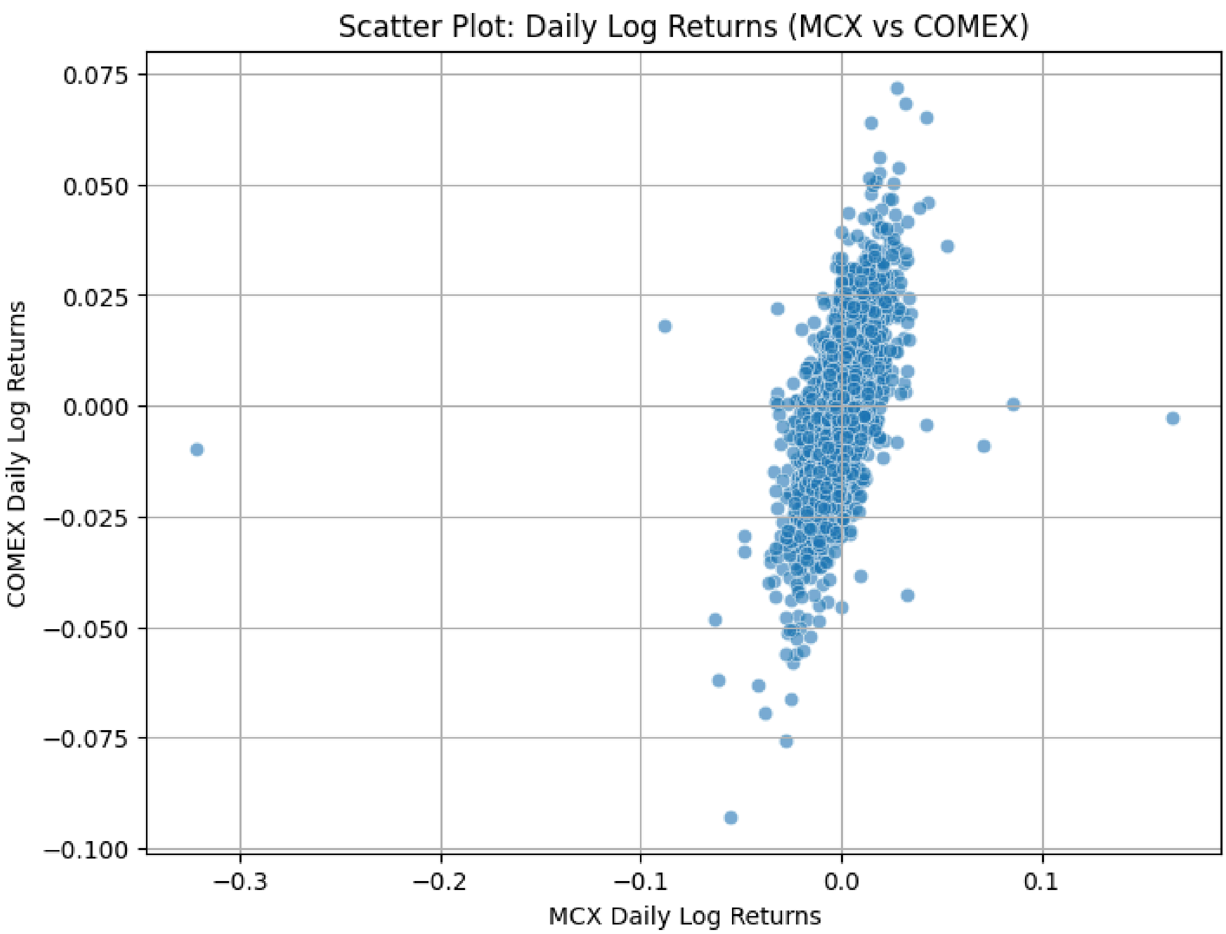
- Codelco's production dropped from 1,672 kMTF in 2014 to 1,325 kMTF in 2023, with production costs rising over 58% to US¢365.7/lb, reflecting operational and profitability pressures.
- Total direct GHG emissions from Chilean copper mining rose from 5.7 million to 6.4 million metric tons of CO_2e (2014–2022), while indirect emissions (electricity use) declined from 14.0 million to 10.7 million metric tons, reflecting some efficiency improvements.

Demand

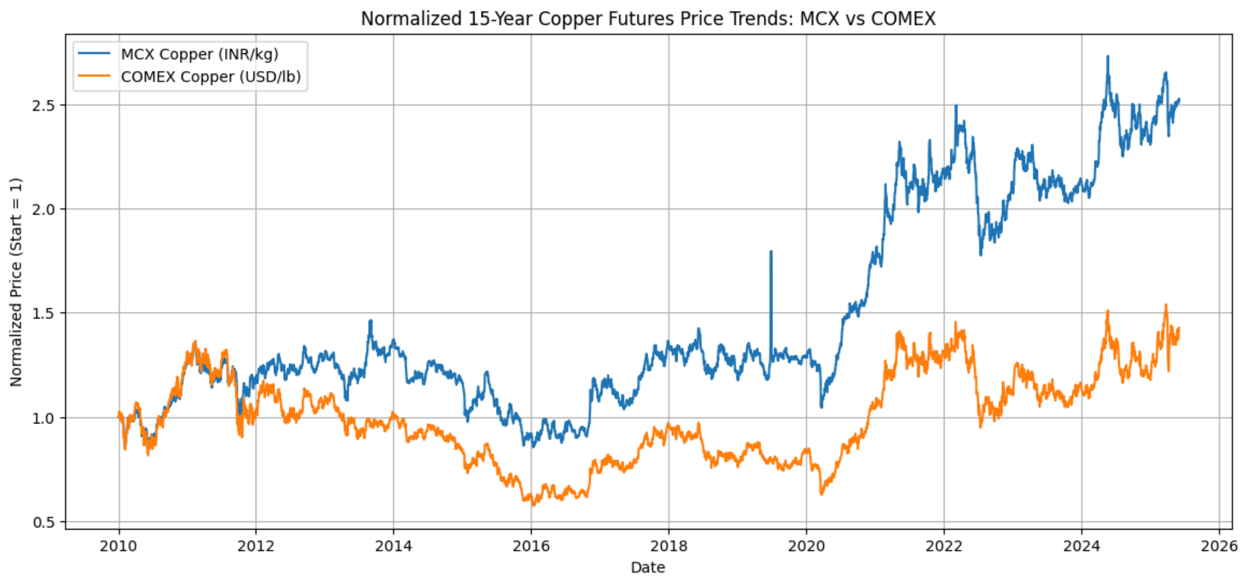
1. *2025 Copper Squeeze:* LME inventories fell 80%, causing COMEX-LME premiums to spike ($1,500/ton) and record backwardation ($345/ton)—a sign of market dislocation from U.S. tariffs and China's raw material crunch. (5) (6)
2. **Current Market Dynamics**: (7)
   a. The copper rally in 2025 is driven more by speculative flows (e.g., funds and traders buying U.S.-priced copper futures and options as a hedge against tariffs) than by strong underlying demand.
   b. Increased demand from Chinese buyers post-Lunar New Year has supported market momentum, despite the rally being less tied to an economic rebound.
   c. Investors are cautioned that copper prices may not always reflect economic health. Temporary shortages can drive prices up even during global economic slowdowns, while a copper glut may lower prices despite robust growth.
   d. The Copper Development Association (CDA) estimates global copper consumption by sector: 46% building construction, 21% electrical, 16% transportation, and 17% consumer products and industrial machinery/equipment.
   e. U.S.: Consumption down ~25% (2003 to 2023) - (1)
   f. Total alloying metal consumption (zinc, lead, tin, nickel) in the U.S. fell from 293 to 135 thousand short tons, indicating a broader decline in metallic industrial inputs. - (1)
   g. Copper prices have risen 8.79% over the past month and approximately 27.14%year-to-date, reaching $5.06 per pound. (8)

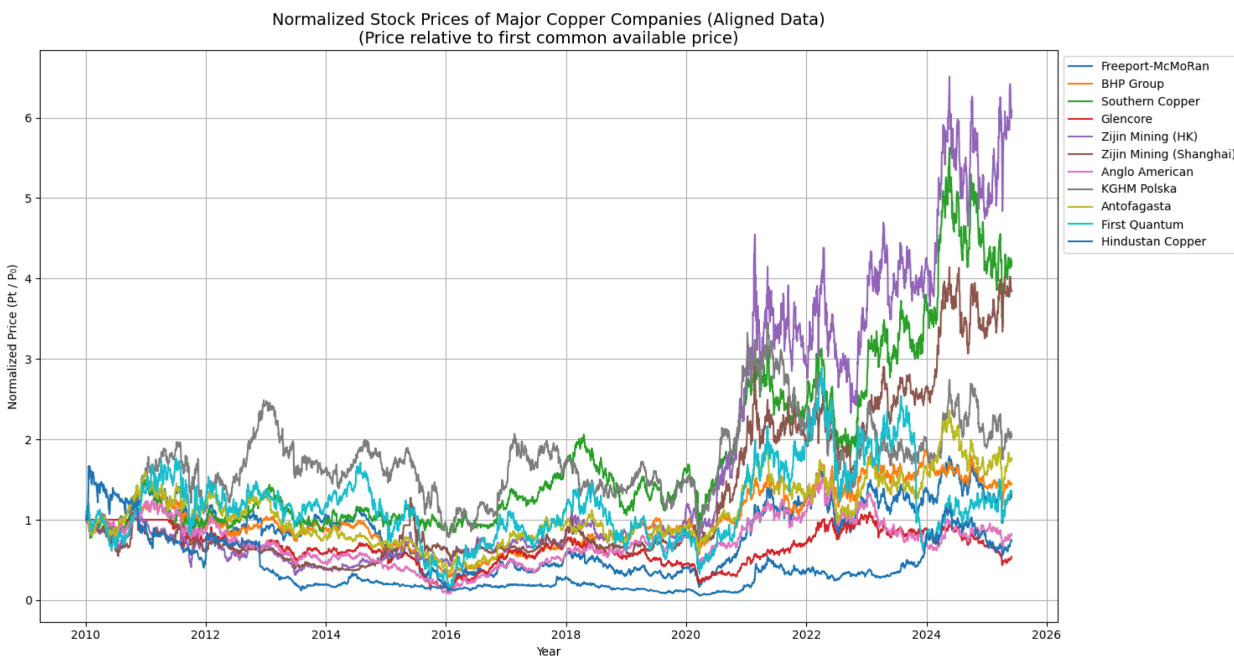
Noteworthy Industry Signals -

1. Hindustan Copper Limited anticipates increased copper demand due to renewable energy and electric vehicles. HCL is exploring new blocks in Chhattisgarh, Rajasthan, and Jharkhand. The company targets a 15-20% rise in copper ore production by 2025-26, aiming for around 4 million tonnes. HCL reported a consolidated net profit of ₹467.42 crore for 2024-25. **(9)**
2. HCL Production Trend (FY 2024–25): MIC: 6587↑(+10.2%) → 6034↓(–9.6%) → 6340↓(–3.8%) **(10)**
3. Freeport-McMor+Ran - 5Y total return = 316.15% **(11)**
4. Southern Copper(SCCO) - 5Y total return = 245.49% **(12)**



When COMEX copper prices move up or down, MCX copper tends to move in the same direction about 57% of the time in a linear sense.


Normalized 15-Year Copper Futures Price Trends: MCX vs COMEX
MCX Copper (INR/kg)
COMEX Copper (USD/lb)
Normalized Price (Start = 1)
Date
2010
2012
2014
2016
2018
2020
2022
2024
2026
0.5
1.0
1.5
2.0
2.5


Normalized Stock Prices of Major Copper Companies (Aligned Data)
(Price relative to first common available price)
Freeport-McMoRan
BHP Group
Southern Copper
Glencore
Zijin Mining (HK)
Zijin Mining (Shanghai)
Anglo American
KGHM Polska
Antofagasta
First Quantum
Hindustan Copper
Normalized Price (Pt / P0)
Year
2010
2012
2014
2016
2018
2020
2022
2024
2026
0
1
2
3
4
5
6